

BVetter Revised Capstone Documentation

Updated based on the system's test_eval.py model evaluation results.

This revision keeps the explanation simple and corrects the model descriptions so they match the current BVetter analytics pipeline.

What Was Changed From the Previous Version

The previous document described some model parts too broadly. This version clarifies which parts are real machine learning models, which parts are ARIMA/SARIMA forecasts, and which parts are rule-based decisions.

Most importantly, the barangay action tier is not a Random Forest prediction. It is now explained as a documented rule based on observed cases. This change is important because the tested classifier only reached 55.9% accuracy, which was worse than the 65.4% majority baseline.

Evaluation Summary

Evaluation item	Result	Simple meaning
Municipality disease forecast	Seasonal ARIMA, about 1.2% MAPE	City-wide disease trend is forecastable.
Barangay disease forecast	Top-down SARIMA allocation	Barangay values are shares of the city-wide forecast.
Barangay action tier	Rule, not ML model	Needs Action, Watch, and Normal use observed cases.
Removed action-tier classifier	55.9% vs 65.4% baseline	The ML classifier was worse than a simple baseline.
Differential diagnosis	RF Classifier: 57.0% top-1, 95.4% top-3	The correct answer may appear in the top 3 suggestions.
Per-disease forecast	Pooled RF Regressor MAE 0.232 vs 0.549 baseline	Pooling diseases gives better case-count estimates.
Disease-specific ARIMA/SARIMA	Avg MAE 1.37 cases/month	Forecasts are usually off by around 1-2 cases.
Vaccination forecast	ARIMA, about 399.6 animals/month	Demand is estimated from municipal historical records.
Pipeline checks	All pass	Totals, shares, intervals, and tier labels are consistent.

Q1. How were paper-based records changed into a centralized system?

Paper-based consultation, vaccination, and disease records were encoded, cleaned, and organized into structured digital data. The system stores operational data in MySQL and uses historical consultation and vaccination datasets for analytics.

In simple terms, the records were changed from scattered manual files into organized data that the system can search, analyze, and use for forecasting.

Q2. How does the system use forecasting and machine learning?

The system uses different models for different jobs. Seasonal ARIMA/SARIMA is used to forecast disease case totals and vaccination demand over time. A Random Forest Classifier is used for differential diagnosis, which means suggesting possible diagnoses from symptoms. A pooled Random Forest Regressor is used for per-disease case-count forecasting across all diseases.

The system does not use a Random Forest model to decide barangay action tiers. The action tier is a rule based on observed case levels, such as whether a barangay is above its usual p75 or p90 case threshold.

Figure 2 — Each Model Against the Baseline It Has to Beat

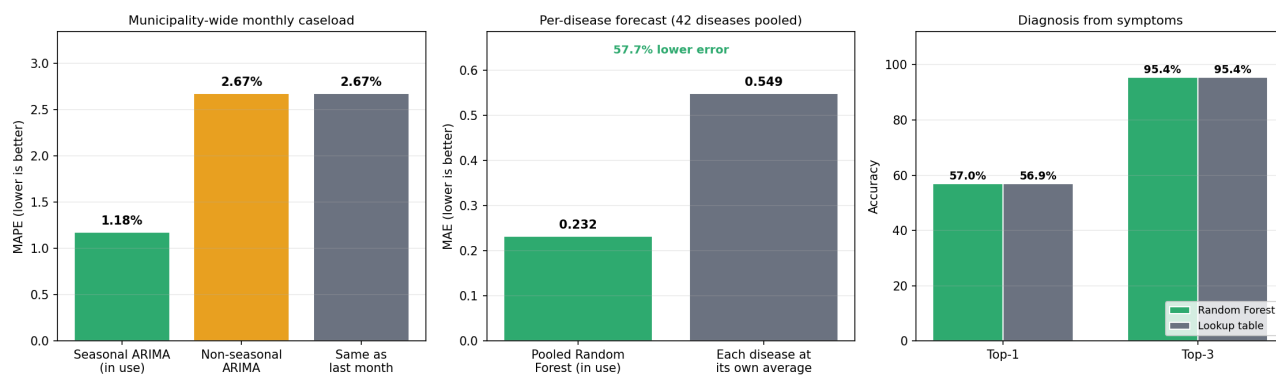


Figure: Models compared with their baselines.

Discussion: The graph shows that Seasonal ARIMA performs better than non-seasonal ARIMA for the municipality-wide forecast. It also shows that the pooled Random Forest Regressor performs better than the simple baseline for per-disease forecasting. For diagnosis, the Random Forest Classifier is strongest when the system shows the top 3 possible diagnoses instead of relying only on the first result.

Q2.1. How are barangay disease risks explained now?

The evaluation showed that individual barangay monthly case counts are too noisy for a reliable separate forecast model. A single barangay has very low lag-1 autocorrelation, about 0.018, meaning this month's value does not strongly predict next month's value.

Because of this, the system forecasts the municipality-wide caseload first using Seasonal ARIMA, then divides that forecast among barangays using their historical share. This is called a top-down approach.

Simple explanation: the system trusts the city-wide trend more than separate barangay trends, because the city-wide data is more stable.

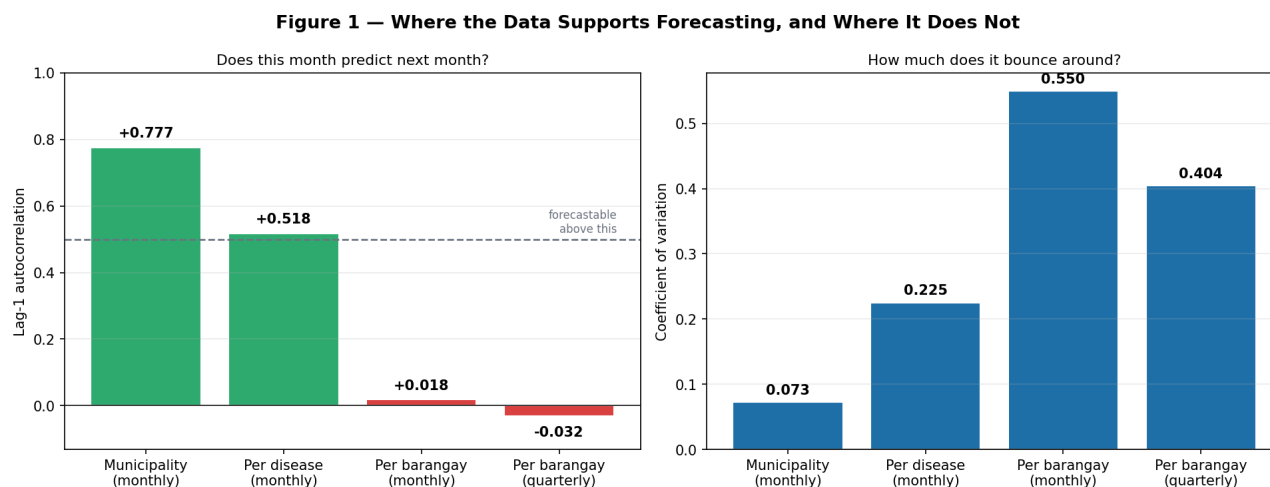


Figure: Municipality and disease-level data are more forecastable than individual barangay monthly data.

Discussion: The result explains why the system does not create separate monthly forecast models for every barangay. Municipality-wide data has a stronger pattern, while individual barangay data changes too much from month to month. Because of this, the top-down method is more appropriate.

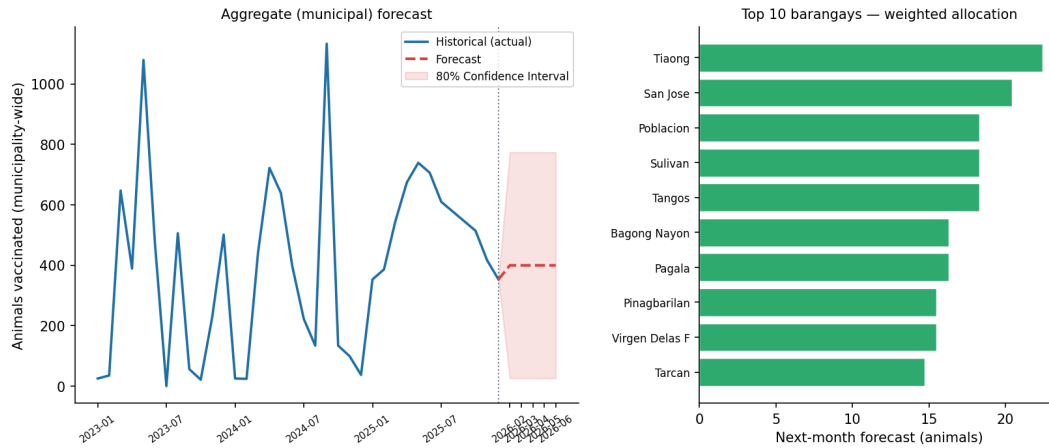
Q2.2. How does ARIMA help with vaccination planning?

The vaccination forecast uses ARIMA on municipality-wide vaccination records. The test evaluation estimated around 399.6 vaccinated animals per month, with the trend marked as stable.

The barangay breakdown is not 27 separate ARIMA models. It is an allocation of one municipal forecast using barangay population or allocation weights.

Simple explanation: BVetter first estimates total vaccine demand, then divides that expected demand per barangay for planning stock and manpower.

Figure 4 — Mass Vaccination ARIMA Forecast (Total Animals Vaccinated) [Model: ARIMA]



Explanation: Left panel — municipality-wide ARIMA forecast from Forecast_Input_* sheets (dogs + cats vaccinated, 36 months). Model: ARIMA. Right panel — the single municipal forecast's next-month value, distributed across the 10 highest-weighted barangays using Barangay_Masterlist's allocation_weight (2025 estimated dog population share). This is a real, documented weighting applied to a real aggregate forecast — not an independently-fit per-barangay time series, since no barangay-level vaccination event history exists in the source data.

Figure: Mass vaccination ARIMA forecast and top barangay allocations.

Discussion: The graph shows a stable vaccination demand of about 399.6 animals per month. The barangay bar chart is not a separate forecast for each barangay; it is the distribution of the municipal forecast based on barangay allocation weights. This helps the clinic plan vaccine supply and manpower.

Q3. How does the system generate insights for decision makers?

The system generates dashboards, charts, barangay summaries, forecast panels, and downloadable reports. These outputs help veterinarians and administrators see disease trends, expected vaccination needs, and areas that may need attention.

The evaluation also checks that the forecast pipeline is internally consistent. Barangay forecasts must add up to the municipal forecast, barangay shares must total 1.0, intervals must contain the forecast value, and the action tier must be shown as a rule without fake model confidence.

In the latest test evaluation, all these invariant checks passed.

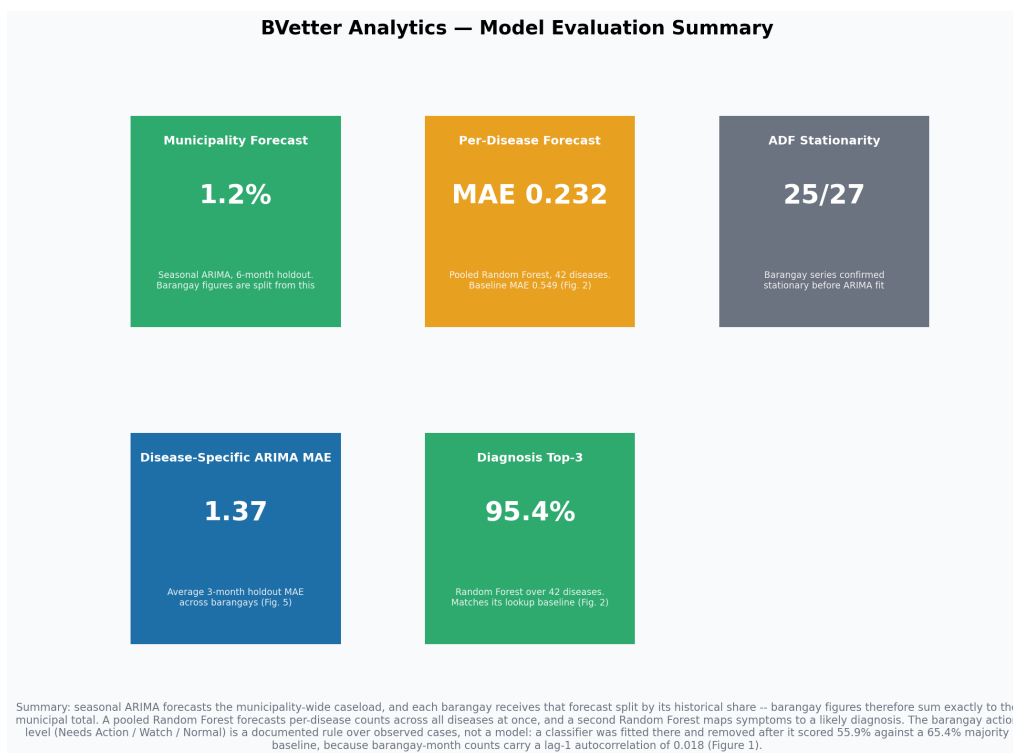


Figure: Overall analytics evaluation summary.

Discussion: The summary shows that the main evaluation results are acceptable to good. The strongest result is the municipality forecast. The barangay action tier is correctly shown as a rule, and the pipeline checks passed, meaning the forecast totals and barangay shares are consistent.

Q4. How can the performance of the implemented models be evaluated?

The performance of the implemented models can be evaluated by comparing their predictions with actual historical data and by checking whether each model performs better than a simple baseline. The system uses different metrics depending on the type of model.

For classification models, the system uses accuracy, precision, recall, and F1-score. Accuracy shows how many predictions are correct overall. Precision shows how many predicted cases are actually correct. Recall shows how many real cases the model was able to identify. F1-score balances precision and recall, which is useful when the classes are not evenly distributed.

For forecasting and regression models, the system uses MAE, MAPE, and RMSE. MAE shows the average error in number of cases. MAPE shows the error as a percentage, which makes it easier to understand. RMSE gives a higher penalty to large errors, so it helps identify forecasts that miss badly.

Based on test_eval.py, the municipality-level Seasonal ARIMA result is good because it reached about 1.2% MAPE. The pooled Random Forest Regressor for per-disease forecasting is also good because it reduced MAE from 0.549 baseline

to 0.232. The diagnosis Random Forest Classifier is acceptable because top-1 accuracy is 57.0%, while top-3 accuracy is 95.4%. This means the first answer is not always correct, but the correct diagnosis is often included in the top three suggestions.

The evaluation also showed that the old barangay action-tier classifier was not reliable because it scored 55.9%, which is lower than the 65.4% majority baseline. Because of this, the system uses a rule-based action tier instead of presenting it as a machine learning prediction.

In simple terms, the results are okay to good. The forecasting models are strong at the municipality level, the per-disease model performs better than baseline, and the diagnosis model is useful when viewed as a top-3 recommendation. However, barangay-level action tier prediction is better handled by clear rules because the machine learning classifier was not reliable enough.

Regression Metrics Based on the Current Evaluation

Pooled Random Forest Regressor: The current Random Forest Regressor is used for per-disease case-count forecasting, not for barangay risk classification. In the latest evaluation, it reached an MAE of 0.232 compared with the baseline MAE of 0.549. This means the model has lower average error than simply using each disease's average value.

All-disease ARIMA/SARIMA forecast: The all-disease forecast is powered by ARIMA/SARIMA. The latest evaluation reported MAE of 2.58, RMSE of 3.89, and MAPE of 1.2%. This is a good result for the municipality-level forecast because the percentage error is low.

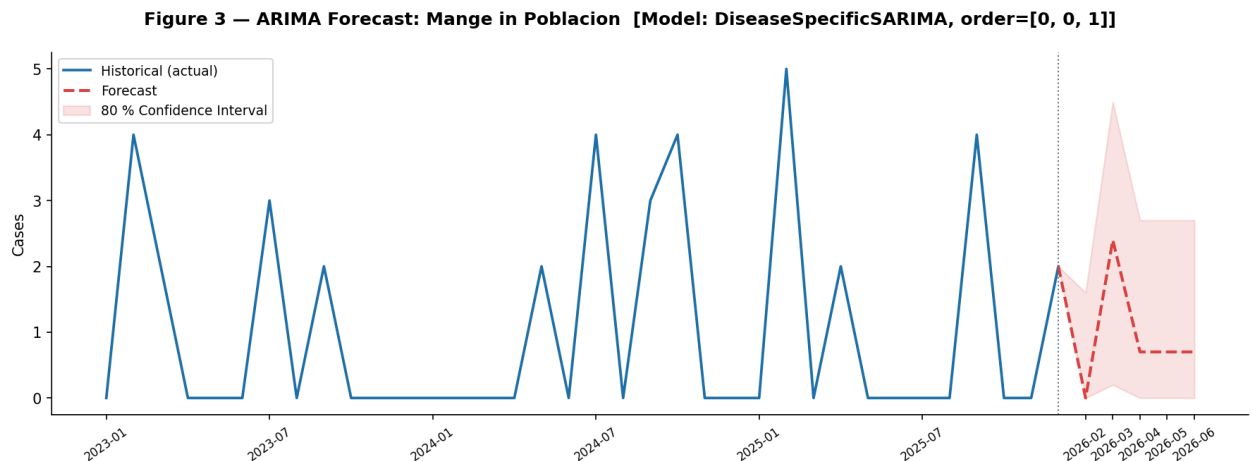
Disease-specific ARIMA/SARIMA forecast: For disease-specific barangay evaluation, the average MAE is 1.37 cases per month and the average RMSE is 1.81. This means the model is usually off by around one to two cases per month. It is useful for planning, but barangays with higher errors should still be checked by clinic personnel.

Mass vaccination ARIMA forecast: The current evaluation no longer uses the older high-error vaccination result. The vaccination data was corrected by treating 2023 and 2024 as year-to-date totals and converting them back into monthly values before fitting ARIMA. After correction, the forecast is stable at about 399.6 animals per month. The barangay values are weighted allocations of the municipal forecast, not separate barangay-level ARIMA models.

The older values such as RF MAE 0.1142 and vaccination ARIMA MAPE 803.1% should not be used in the revised explanation because they describe an older or uncorrected evaluation. The current document should use the latest test_eval.py results instead.

Q4.1. What do the evaluation figures mean?

Figure 3: shows a sample ARIMA forecast for Mange in Poblacion. The forecast trend is stable, with sample forecast values of 0.0, 2.4, 0.7, 0.7, 0.7, and 0.0 cases.



Explanation: The blue line shows actual monthly case counts for Mange in Poblacion. The dashed red line is the DiseaseSpecificSARIMA forecast for the next 6 months. The shaded band is the 80 % prediction interval — narrower bands indicate more confident forecasts. Trend detected: Stable. ARIMA uses past values and error terms to extrapolate the time series pattern.

Figure: Sample ARIMA forecast for a disease and barangay.

Discussion: The sample graph for Mange in Poblacion shows a stable trend. The forecast values are low, which means no major sudden increase is expected in this sample. The shaded interval shows the possible range of future values, so users should read the forecast as an estimate, not an exact number.

Figure 5: shows disease-specific ARIMA/SARIMA holdout errors across barangays. The average MAE is 1.37 cases per month and the average RMSE is 1.81.

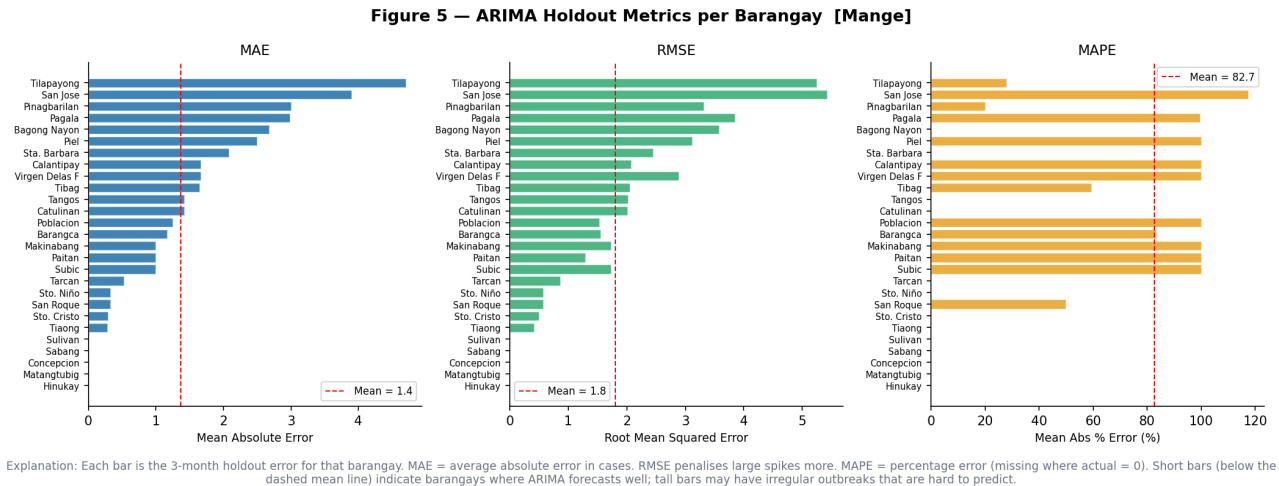


Figure: Disease-specific ARIMA/SARIMA error metrics by barangay.

Discussion: The graph compares forecast error across barangays. Lower bars mean better prediction performance. The average error is about 1.37 cases per month, which is acceptable for planning, but barangays with higher error should still be reviewed by clinic personnel.

Q4 Continued. ARIMA Validation

The residual and stationarity checks help verify whether ARIMA is behaving properly. The sample ADF test result had a p-value of 0.0, meaning the sample series was stationary. Across barangays for the evaluated disease, 25 of 27 series were already stationary.

Simple explanation: ARIMA works better when the data pattern is stable enough. The ADF test helps the system decide whether differencing is needed before fitting the model.

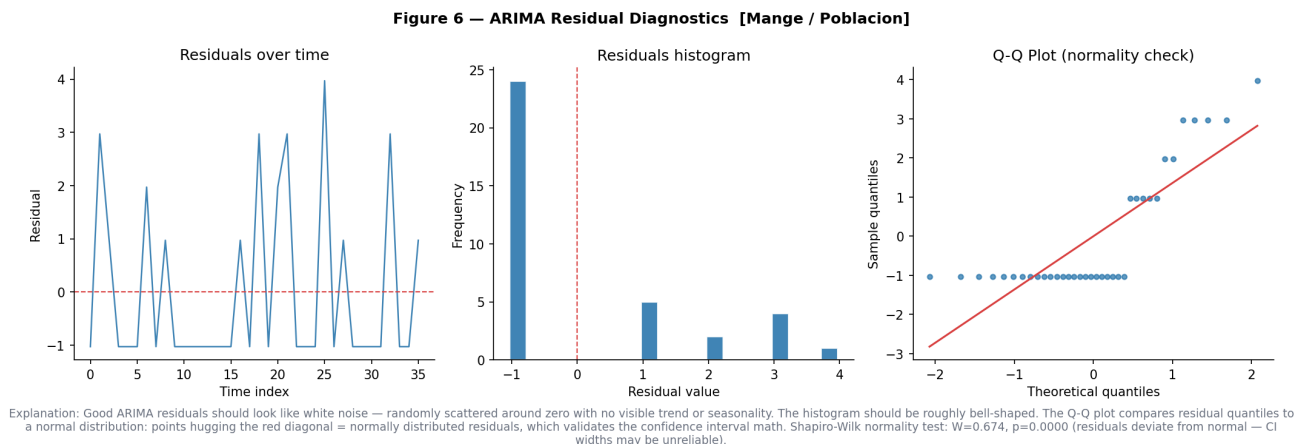


Figure: ARIMA residual diagnostics.

Discussion: The residual graph checks if the ARIMA errors behave randomly. The Shapiro-Wilk result showed the residuals are not perfectly normal, so the confidence intervals should be interpreted carefully. Still, this diagnostic is useful because it tells users where the model may have remaining uncertainty.

Q5. How do the chatbot and lost-and-found module improve the system?

The chatbot improves access to veterinary information by answering common questions and giving basic symptom-based recommendations. The lost-and-found module improves recovery by matching reports based on species, breed, size, color, barangay/location, notes, and uploaded images.

These modules are not part of test_eval.py, but they remain important system features because they support public service, faster information access, and easier lost-pet recovery.

Final Simple Summary

The updated system explanation should say: BVetter uses Seasonal ARIMA/SARIMA for time-series forecasting, Random Forest for diagnosis and pooled per-disease forecasting, and simple documented rules for barangay action tiers. The evaluation supports this design because municipality-level data is more predictable than individual barangay monthly data.